

EXTERNAL MEMORANDUM

Logos, Pathos & Ethos

A concise guide to reasoning and persuasion

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Aristotle described three main ways people persuade one another.¹ These are useful for analysing speeches and advertising, learning how to make a sound argument and recognizing attempts to influence us.

¹ In Aristotle's *Rhetoric*, logos, pathos and ethos are distinct ways of making a case, but effective persuasion commonly combines all three.

	LOGOS	PATHOS	ETHOS
Meaning	Logic and reason	Emotion	Credibility and character
Main question	Does this make sense?	How is this making me feel?	Why should I trust this person?
Common tools	Facts, evidence, examples, cause and effect	Fear, sympathy, humour, pride, anger and hope	Expertise, experience, reputation and honesty
Weakness	Evidence can be selective or misleading	Feeling can overwhelm judgment	Authority can be false or irrelevant

LOGOS *The argument*

Logos is an appeal to **reason**: it asks whether the conclusion follows from the evidence.²

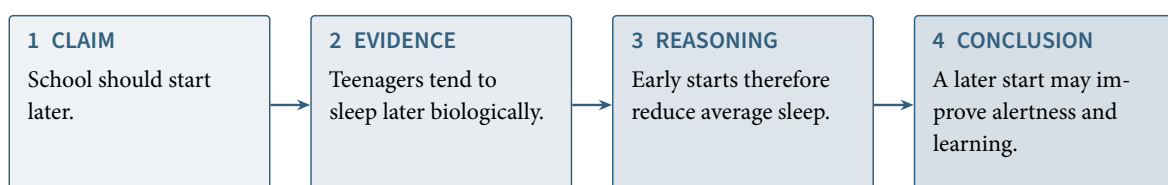
'School should start later: teenagers tend to sleep later biologically, so early starts reduce sleep and may impair learning.'

² A logically valid argument can still have a false conclusion if one of its premises is false. Good reasoning therefore tests both the structure of the argument and the truth of its evidence.

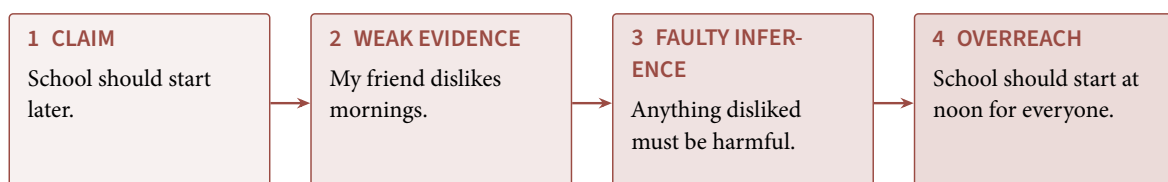
QUESTIONS TO ASK

1. What exactly is the claim?
2. What evidence supports it?
3. Does the evidence actually support the conclusion?
4. Are there other possible explanations?
5. Is anything important being left out?
6. What evidence would change my mind?

A SUCCESSFUL REASONING CHAIN



AN UNSUCCESSFUL REASONING CHAIN



PATHOS *The feeling*

Pathos is an appeal to **emotion**.

‘Look at these sad and hungry abandoned animals currently at our animal shelter.’

Emotion is not automatically dishonest. It is an important part of human judgment. The problem occurs when emotion is used *instead of evidence*.³

³ Psychologists sometimes call this the *affect heuristic*: our immediate positive or negative feeling can become a shortcut for judging risk, truth or value. The feeling may be informative, but it is not proof.

TWO TESTS FOR EMOTIONAL APPEALS

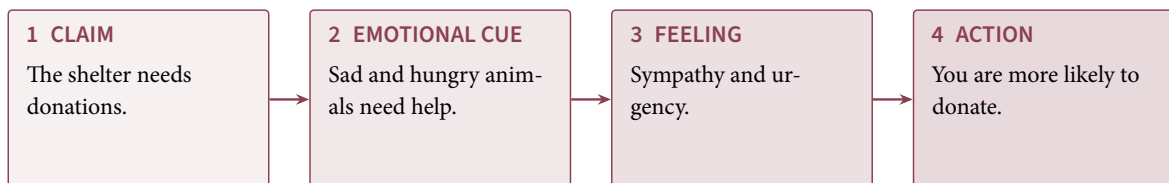
1. What emotion is this trying to produce in me?

fear	anger	sympathy
disgust	excitement	pride
belonging	envy	humour

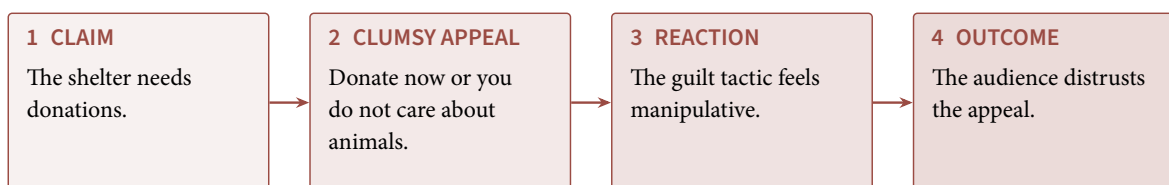
2. Would the argument remain convincing if I were not feeling that emotion?

Advertising and political rhetoric frequently become easier to analyse once this question is asked.

A SUCCESSFUL REASONING CHAIN



AN UNSUCCESSFUL REASONING CHAIN



ETHOS *The speaker*

Ethos concerns why we should believe the person making the argument.

‘I have repaired this type of engine for thirty years.’

Relevant expertise is sensible evidence, but authority needs examining.⁴

QUESTIONS TO ASK

1. Who is speaking?
2. What relevant expertise do they have?
3. How do we know they are reliable?
4. Might they have something to gain?⁵
5. Are they speaking within their field of expertise?
6. Can their claims be independently checked?

WHO BENEFITS? WHO BEARS THE RISK?

A speaker may benefit from persuading you while remaining untouched by the consequences. Ask whether their interests align with yours, whether they bear any cost if their advice is wrong, and who ultimately carries the risk.

‘This person is trustworthy’ is not the same as ‘this claim is true.’

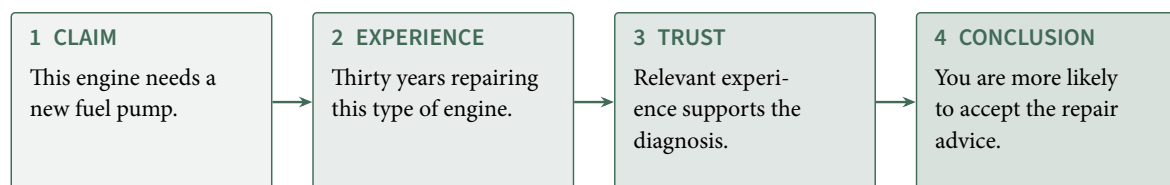
Experts can be wrong.⁶ Ask to see the evidence and reasoning: a CV is not evidence for this particular claim.

⁴ Expertise is domain-specific. A person may be highly reliable in one field and no better informed than anyone else in another. Independence, transparency and a relevant track record also matter.

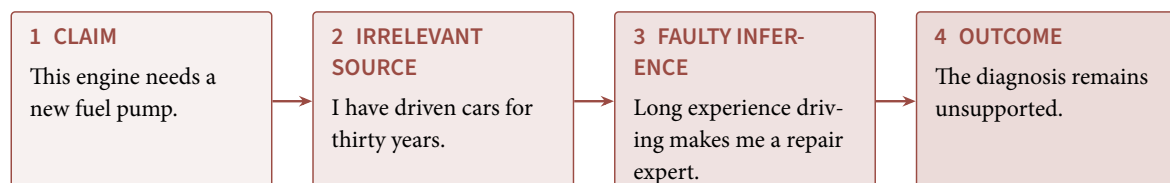
⁵ **Follow the incentives.** Someone need not care what happens to you in order to sound confident, sympathetic or trustworthy.

⁶ I find that a good response to the appeal to authority argument begins with something like this: “Well, then it should be extremely easy for you to demonstrate. Name the <claim you are making> in <subject under discussion> and explain how it comes about, and how we laypersons can see it. Your CV is not evidence.”

A SUCCESSFUL REASONING CHAIN



AN UNSUCCESSFUL REASONING CHAIN



A PRACTICAL EXERCISE *What is missing?*

THREE WAYS IN, ONE WAY OUT

Persuasion enters through reason, emotion and trust; critical inquiry provides the exit. Take the nearest advert, speech, TikTok, newspaper headline or argument you can eavesdrop on and mark its components:

In

1. **LOGOS:** What reasons or evidence are offered?
2. **PATHOS:** What am I encouraged to feel?
3. **ETHOS:** Why am I encouraged to trust the speaker?

Out

1. **MISSING:** What information would help me judge the claim?

TRY IT ON A FAMILIAR CLAIM

Advert	'Nine out of ten dentists recommend Brand X.'
Ethos	Dentists are presented as experts.
Logos	The claim appears to offer statistical evidence.
Pathos	'Surely, comrades, you do not want to neglect your teeth and be the laughing stock of your friends?'

NOW INTERROGATE WHAT IS ABSENT

The most useful questions often concern the information that was not supplied: ⁷

1. Nine out of ten **of how many dentists?**
2. What exactly were they asked?
3. Who paid for the survey?
4. **Who benefits, and who bears the risk?**
5. Did they recommend other brands too?

THE LESSON TO REMEMBER

The goal is not to teach 'Logos good; Pathos bad; Ethos suspicious.' Strong arguments usually contain **all three**.

Separate **what you are being asked to believe**, **how you are being made to feel**, and **why you are being asked to trust the person** saying it.

Once someone learns to separate those three things, they become considerably harder to manipulate.

⁷ A percentage without its denominator can be deceptive. Nine out of ten is strong evidence if ten people were sampled carefully, but weak evidence if respondents were selected or the question was framed to invite agreement. And that's before we get into statistical significance and small sample sizes. Small sample sizes are the scourge of arguments - you could easily argue that tortoises are faster than hares based on a single race. Try that over 200 races.

PRACTICE *From recognition to judgment*

Work through these exercises in order. The early questions have a best answer. The later ones require judgment because facts are incomplete, values conflict, and every available decision imposes risks on someone.

Use the same 3+1 test throughout: **What reasons are offered? What emotion is being encouraged? Why should the source be trusted? What is missing?**

LEVEL 1 — IDENTIFY THE APPEAL

Choose the best answer.

1. 'A randomised trial assigned 2,000 patients to two treatments. Recovery was eight percentage points higher with Treatment A.' Which appeal is primary?
 - A. Logos
 - B. Pathos
 - C. Ethos
 - D. None of the three
2. 'Imagine your child waiting alone in a cold, dark station because the last bus was cancelled.' Which appeal is primary?
 - A. Logos
 - B. Pathos
 - C. Ethos
 - D. Statistical inference
3. 'I have inspected suspension bridges for thirty years, and this crack requires immediate attention.' Which appeal is primary?
 - A. Logos
 - B. Pathos
 - C. Ethos
 - D. Popularity
4. A famous actor recommends a heart medicine. What should be checked first?
 - A. How many followers the actor has
 - B. Whether the actor has relevant expertise and a financial incentive
 - C. Whether the advert is emotionally moving
 - D. Whether the actor appears confident
5. A council reports that complaints fell from four to two after a new policy. Which missing fact matters most?
 - A. The colour of the report
 - B. Whether the periods, populations and reporting methods are comparable
 - C. Whether the mayor announced the result

- D. Whether the policy has an attractive name
- 6. 'Nine out of ten dentists recommend Brand X' combines which two appeals most directly?
 - A. Logos and ethos
 - B. Logos and pathos
 - C. Pathos and ethos
 - D. Pathos and humour

PRACTICE *Diagnosing weak arguments*

LEVEL 2 — NAME THE FAULT

Choose the most important defect in each argument.

1. 'Either approve this safety bill today or admit that you do not care whether children live or die.'
 - A. False dilemma reinforced by pathos
 - B. Valid deduction
 - C. Representative sampling
 - D. Relevant authority
2. 'The town installed cameras in March. Crime fell in April. Therefore the cameras caused the fall.'
 - A. Equivocation
 - B. Treating sequence or correlation as proof of causation
 - C. Appeal to pity
 - D. Deduction from a universal rule
3. 'Three constituents wrote to oppose the proposal. The public clearly rejects it.'
 - A. Hasty generalisation from a self-selected sample
 - B. Modus tollens
 - C. Appeal to relevant expertise
 - D. A controlled experiment
4. 'A Nobel Prize-winning physicist says this tax plan will increase wages, so it must be true.'
 - A. The authority may be prestigious but irrelevant to the claim
 - B. Physics can never inform policy
 - C. Experts are always wrong outside laboratories
 - D. Tax plans cannot be evaluated
5. A treatment increases recovery from 1% to 2%. An advert says it 'doubles the chance of recovery' but omits the absolute figures. What is the main problem?
 - A. The relative change is false
 - B. The framing conceals the base rate and absolute effect

- C. Percentages are never evidence
- D. The sample must contain exactly one hundred people

LEVEL 3 — LOGIC AND EVIDENCE PUZZLES

1. A reliable database follows this rule: *if a permit is valid, its record is green*. Sam's record is not green. What follows?
 - A. Sam's permit is not valid
 - B. Sam has never applied for a permit
 - C. Every green permit belongs to Sam
 - D. Nothing follows
2. *Only licensed drivers may operate the school bus*. Mira has a licence. What may we conclude?
 - A. Mira operates the school bus
 - B. Mira is permitted by this rule, but may not actually operate the bus
 - C. Everyone with a licence operates a school bus
 - D. Mira is employed by the school
3. A survey conducted at a demonstration finds that 82% of respondents oppose the policy being protested. What is the central difficulty?
 - A. The percentage is too precise
 - B. The sample is strongly selected and may not represent the public
 - C. Demonstrators are prohibited from answering surveys
 - D. A sample must include the entire population
4. All emergency orders require either legislative approval or a declared disaster. An order has no legislative approval, but a disaster has been declared. Is the order necessarily lawful?
 - A. Yes; satisfying one stated requirement proves full legality
 - B. No; the rule states a necessary procedural condition, not every condition of legality
 - C. Yes; emergency action cannot be reviewed
 - D. No; declarations of disaster are never valid

A CASE-MIX PUZZLE

Two hospitals report the following thirty-day survival figures:

Hospital	Routine cases	High-risk cases	All cases
Hospital A	90/100 (90%)	30/50 (60%)	120/150 (80%)
Hospital B	19/20 (95%)	56/80 (70%)	75/100 (75%)

Hospital A has the better overall rate, while Hospital B has the better rate within *each* risk category. What should an investigator conclude?

- A. Hospital A is unambiguously better
- B. Hospital B is unambiguously better in every relevant respect
- C. The overall figures alone settle the question
- D. Case mix changes the aggregate result; comparison requires risk adjustment and more evidence

PRACTICE *Mixed appeals and incomplete stories*

LEVEL 4 — ANNOTATE, CHALLENGE AND REPAIR

For each scenario:

1. mark the logos, pathos and ethos;
2. state the principal claim in neutral language;
3. identify at least three missing facts or perspectives;
4. rewrite the argument so that its evidence and conclusion are proportionate; and
5. state what evidence would change your mind.

SCENARIO A — THE BUS LANE

A traders' association shows photographs of three empty shops and says: 'The new bus lane is destroying our high street. Remove it before the whole district dies.' It reports that sales fell 12% after installation. The association does not show city-wide retail trends, pedestrian counts, rents, online sales, construction disruption or results from comparable streets.

SCENARIO B — THE FOUNDER

A technology company's founder tells legislators: 'I built this industry, so I know that regulation will destroy innovation.' The company funds the study he cites. Independent researchers agree that compliance has costs but disagree about their size and whether benefits exceed them.

SCENARIO C — THE VIRAL CLIP

A twenty-second video of a frightening incident outside a school is viewed ten million times. A campaign demands permanent police deployment at every school. The clip is genuine, but no one provides the frequency of such incidents, comparisons with other risks, evidence about alternative safeguards, or the effects of permanent policing on different pupils.

SCENARIO D — THE SUCCESS RATE

A minister announces that a job programme is 'proven to work' because 70% of participants found employment. Participation was voluntary, 68% of eligible non-participants also found work, and the two groups differed in age, health and prior employment. The programme may still help, but the headline figure does not establish how much.

PRACTICE *Decisions with no clean answer*
LEVEL 5 — LAW, GOVERNMENT AND CONFLICTING VALUES

These cases are difficult because reasonable people can share the same facts and still rank rights, risks and obligations differently. For each case:

1. separate empirical claims from moral, legal and political judgments;
2. give the strongest logos, pathos and ethos available to opposing sides;
3. identify incentives, conflicts of interest and absent stakeholders;
4. ask who benefits, who bears the risk and who can appeal;
5. propose a decision with safeguards, review criteria and a route for correction; and
6. state the strongest objection to your own proposal.

CASE 1 — PRE-TRIAL DETENTION

After several serious offences allegedly committed by people awaiting trial, a city proposes an algorithm to recommend detention without bail for those rated high risk. Officials say retrospective testing predicts rearrest better than judges alone. Civil-rights groups note that historical arrest data reflect unequal policing, detention can cost innocent people jobs and homes, and the vendor will not disclose the model. Releasing someone who causes grave harm and detaining someone who would not offend are both irreversible failures borne by different people.

CASE 2 — ENCRYPTION AND LAWFUL ACCESS

After a deadly attack, investigators seek a legal power requiring messaging services to provide exceptional access to encrypted conversations when a court issues a warrant. Families argue that inaccessible evidence leaves victims without justice. Security engineers argue that an access mechanism usable by authorities may also be discovered or abused by criminals and hostile states. The government relies partly on classified evidence; companies benefit from a reputation for privacy; no design has been shown to eliminate both risks.

PRACTICE CONTINUED *Institutional dilemmas*
CASE 3 — WATER IN A SEVERE DROUGHT

A reservoir is at 30% capacity. Long-standing law gives some farms senior rights; cities need drinking water; a treaty protects an Indigenous nation's share; minimum flows support fisheries and an endangered ecosystem. Cutting farms may bankrupt families and disrupt food supply. Cutting cities threatens health. Cutting river flows may cause irreversible ecological loss. Rainfall forecasts are uncertain, and every group can produce respected experts.

CASE 4 — EMERGENCY POWERS

A fast-moving epidemic reaches several regions. The executive seeks twelve months of authority to restrict movement, redirect public spending and issue rules without prior legislative approval. Delay may cost lives; unchecked power may burden minorities, suppress protest and persist after the emergency. The legislature is slow and divided, courts act retrospectively, and the severity of the threat cannot yet be estimated confidently.

CAPSTONE — DESIGN THE DECISION PROCESS

Choose one of the four cases. Do not begin by choosing the policy outcome. Instead design a process that people who expect to lose could still regard as legitimate. Specify who decides, what evidence is public, which rights cannot be traded away, how minority interests are represented, when the decision expires, what triggers review, and what remedy exists when the policy causes unjustified harm.

The 3+1 test cannot make value conflicts disappear. Its purpose is to expose the argument, the emotional pressure, the claimed authority and the omissions *before* power is exercised.

ANSWER NOTES *Closed questions and open judgments*
LEVEL 1

1. **A — Logos.** The claim relies primarily on comparative evidence.
2. **B — Pathos.** The imagined scene is designed to produce concern.
3. **C — Ethos.** Relevant experience supports credibility.
4. **B.** Fame is not medical expertise; payment or sponsorship may affect independence.
5. **B.** A change from four to two is uninterpretable without comparable exposure, periods and reporting.
6. **A — Logos and ethos.** The number suggests evidence; dentists supply authority.

LEVEL 2

1. **A.** It excludes other positions and uses fear and guilt.
2. **B.** Temporal order does not isolate cameras as the cause.
3. **A.** Three self-selected writers cannot establish public opinion.
4. **A.** Achievement in physics does not settle an economic claim.
5. **B.** The relative claim is numerically true but hides a one-percentage-point absolute increase.

LEVEL 3

1. **A.** Given a reliable rule, this is modus tollens: valid implies green; not green therefore implies not valid.
2. **B.** Licensing is necessary for operating the bus, not sufficient proof that Mira operates it.
3. **B.** The sampling location selects people unusually likely to oppose the policy.
4. **B.** Satisfying the stated either/or condition does not prove compliance with every other legal requirement.
5. **D.** Hospital B has higher survival within both risk groups, while Hospital A treats a much larger share of routine cases. Neither crude total answers every quality question.

ANSWER NOTES CONTINUED *How to assess the open exercises*

LEVEL 4

Good answers distinguish a vivid example from a representative pattern, identify who supplied each statistic, seek comparison groups and time trends, and reduce the certainty of the conclusion when causal evidence is weak. A repair should not simply reverse the original conclusion; it should make a claim no stronger than the evidence permits.

LEVEL 5

The public-law cases have no answer key that can substitute for judgment. Two readers may reach different conclusions and both reason well. Assess the *quality of the decision process* using this rubric:

Criterion	What a strong answer does	Score
Claims and evidence	Separates facts, predictions, causal assumptions and value judgments.	0–2
Uncertainty	States what is unknown and avoids pretending that estimates are certainties.	0–2
Appeals	Identifies logos, pathos and ethos without assuming that emotion or authority is automatically illegitimate.	0–2
Interests and risk	Identifies incentives, affected non-speakers, distributional effects and who bears each kind of error.	0–2
Rights and constraints	Distinguishes outcomes that may be balanced from rights or duties treated as firm limits.	0–2
Safeguards	Uses transparency, reasons, appeals, independent oversight, sunset clauses, thresholds or staged implementation.	0–2
Reversibility	Prefers correctable experiments where possible and treats irreversible harms with greater caution.	0–2
Counterargument	States the strongest objection fairly and changes the proposal in response.	0–2

A strong answer need not reach a particular political conclusion. It must show why its evidence is relevant, which values control the decision, whose interests are at stake, and how mistakes can be detected and corrected.

APPENDIX Words, roots and relationships

The relationships are direct, which makes the three terms easier to remember. The important wrinkle is that their English descendants have sometimes **narrowed or shifted in meaning**.

Greek root	Basic Greek sense	English descendant	What happened
λόγος — logos	word, account, reason, explanation	logic	Retained the sense of reasoning.
ἦθος — ethos	character, disposition, customary behaviour	ethics	Developed into reasoning about good character and conduct.
πάθος — pathos	experience, feeling, suffering	pathetic	Narrowed from arousing emotion to arousing pity, then often to pitiful or contemptible.

LOGOS TO LOGIC

This is the most obvious relationship.

Greek **logos** has a remarkably broad range of meanings: *word, speech, account, explanation, reason, principle*. From it came **logikē**—roughly, *the art concerned with reasoning*—and ultimately the English word **logic**.

Rhetorical logos is not identical to logic, but they are close relatives:

Logos: ‘Here are the reasons you should believe me.’
Logic: ‘Do those reasons actually follow?’

That distinction is useful when teaching. An argument can **appeal to logos while containing bad logic**. Statistics, graphs and apparently rational explanations are still appeals to logos even when they are selective or misleading.

ETHOS TO ETHICS

This relationship is perhaps even more interesting.

Greek **ethos** concerns **character, habit or disposition**: essentially, what sort of person someone is. **Ethikos** means ‘pertaining to character’, which gives us **ethics**.

That makes Aristotle’s rhetorical ethos much clearer than simply translating it as ‘authority’.

The question is not merely ‘Is this person an expert?’
 It is more fundamentally: ‘**What kind of person does the speaker appear to be?**’

Do they seem knowledgeable, honest, fair, sensible, one of us, or to be acting in good faith? Ethics grows from the same territory: questions about **character and conduct**. The word *ethos* also survives unchanged in English, as in ‘the ethos of the organisation’.

APPENDIX CONTINUED *Pathos and its word family*

PATHOS TO PATHETIC

This one has undergone the biggest semantic journey.

Greek **pathos** means something like **that which is experienced or suffered**, and consequently an **emotion, feeling or passion**. This is the source of Aristotle's rhetorical pathos: affecting the emotional state of an audience.

Pathetikos meant *capable of feeling or capable of arousing emotion*. English **pathetic** therefore originally meant something much closer to 'moving; emotionally affecting; arousing pity'. In nineteenth-century writing, 'a pathetic scene' is a moving scene, especially one that evokes pity—not necessarily a bad one.

emotionally affecting → deserving pity → pitiful → feeble or contemptible

Other descendants preserve more of the original connection with feeling:

sympathy *syn* ('with') + *pathos*: feeling with.

empathy *em/en* ('in') + *pathos*: feeling into.

apathy *a-* ('without') + *pathos*: without feeling.

antipathy *anti-* ('against') + *pathos*: feeling against.

psychopathology / pathology Originally concerned with suffering, disease, or what happens to something.

A MNEMONIC TO REMEMBER

LOGOS → LOGIC Think	ETHOS → ETHICS Character	PATHOS → PATHETIC Feel
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Remember: *pathetic* originally meant emotionally moving, not 'useless'.

The etymology supplies a mnemonic rather than asking learners to memorise three arbitrary Greek words.

Logos for the head, pathos for the heart, ethos so nobody thinks you're a charlatan.